

Math 421

Wednesday, October 1

Announcements

- Homework 4 due Friday
- Grace's Drop-In Hours
 - **Today:** 12-1 VV 311
 - Thursday: 9-11 VV B205
- MLC
 - Proof Table: M-Th 3-7:30 VV B227
 - Course Assistant: Th 4-7 Table 4
- Exam 1 – 10/15

Example

Theorem. If $\lim s_n = s$, $s_n \geq 0$ for all n , and $s > 0$, then $\lim \sqrt{s_n} = \sqrt{s}$.

Recall – Discussion.

Proof.

Activity – Fill in the Blanks

Theorem. If $\lim s_n = s$, then $\lim 5s_n = 5s$.

Proof. Fix $\epsilon > 0$. Since $\lim s_n = s$, there exists some N such that: if $n > N$, then $|s_n - s| < \underline{\hspace{2cm}}$.

Assume $n > N$. Then

$$|5s_n - 5s| = \underline{\hspace{10cm}} < \epsilon.$$

Since $\epsilon > 0$ was arbitrary, $\lim 5s_n = 5s$.

Q: What would need to be modified if you wanted to instead to prove $\lim ks_n = ks$, for $k \in \mathbb{R}$?

Activity

Theorem. If $\lim s_n = s$ and $s_n \in [-M, M]$ for all n , then $\lim(s_n)^2 = s^2$.

1. **Scratch Work:** Find an upper bound for $|s_n^2 - s^2|$ in terms of $|s_n - s|$.
 - *Hint:* $a^2 - b^2 = (a + b)(a - b)$.
 - **Reminder:** The only term in the bound that can depend on n is $|s_n - s|$ itself. Everything else must be independent of n .

2. **Prove Theorem.**

Activity

Theorem. If $\lim s_n = s$ and $s_n \in [-M, M]$ for all n , then $\lim(s_n)^2 = s^2$.

Proof.

Divergence

Def: A sequence that does not converge is said to diverge.

Examples:

Discussion Questions

- ❖ Write the negation of the definition of convergence.
- ❖ What would you have to show in order to prove a sequence diverges?
- ❖ Discuss how you might show that $(-1)^n$ diverges.

Choose Your Own Adventure



$$\lim_{n \rightarrow \infty} \frac{(-1)^n}{n} = 0$$

$$\lim_{n \rightarrow \infty} \frac{1}{n^{1/3}} = 0$$



$$\lim_{n \rightarrow \infty} \frac{2n - 1}{3n + 2} = \frac{2}{3}$$

Let (s_n) be a sequence of nonnegative real numbers and suppose $\lim_{n \rightarrow \infty} s_n = 0$. Prove $\lim_{n \rightarrow \infty} \sqrt{s_n} = 0$.



$$\lim_{n \rightarrow \infty} \left(\sqrt{n^2 + 1} - n \right) = 0$$

Prove that $(-1)^n$ diverges.

Limits are Unique

Thm: If $\lim s_n = s$ and $\lim s_n = t$, then $s = t$.

Proof.

Chapter 2: Sequences

Section 9: Limit Theorems for Sequences

Bounded

Definition. A sequence (s_n) is said to be **bounded** if the set $\{s_n : n \in \mathbb{N}\}$ is a bounded set.

Activity –

1. Give an example of a bounded and an unbounded sequence.
2. Does every bounded sequence converge? Does every unbounded sequence diverge?

Theorem 9.1

Thm: Convergent sequences are bounded.

Discussion.

Proof.